GTO and IGCT Thyristors

Device Principles, Switching Behaviour and Gate Driver Design

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Chapter 1

Introduction

The gate turn-off thyristor (GTO) and its direct descendant, the integrated gate-commutated thyristor (IGCT), are the highest-power fully controllable semiconductor switches available in a single die. They occupy the region of the power-electronics landscape above the reach of a single IGBT module: blocking voltages of 4.5 kV to 10 kV and controllable turn-off currents of several kiloamperes, at the cost of a switching frequency limited to a few hundred hertz [1, 2].

Both devices are four-layer, latching, bipolar structures. Their distinguishing feature with respect to the conventional phase-control thyristor is the ability to interrupt the anode current by extracting the stored charge through the gate terminal. What separates the GTO from the IGCT is not the silicon alone but the way in which that extraction is performed — and therefore the gate driver. The IGCT is, by definition, a gate-commutated thyristor (GCT) sold together with its gate unit; the gate unit is part of the device, not an accessory [3, 4].

For this reason the gate driver is treated here as a full chapter (Chapter 4) rather than as an application note. The design of the gate unit determines the maximum controllable current, the failure mode, the need for a turn-off snubber and, ultimately, the cost and volume of the whole converter.

1.1 Scope of the document

Chapter 2 describes the structure and the physical operating principle of the GTO and of the GCT/IGCT. Chapter 3 covers static and dynamic characteristics, losses and safe operating area. Chapter 4 is dedicated to the gate driver: requirements, topologies, sizing equations, layout, auxiliary power supply, monitoring and protection, with a worked numerical example. Chapter 5 summarises typical applications and compares the devices with high-voltage IGBTs. Numerical values quoted throughout the text are representative orders of magnitude; the data sheet of the specific device always takes precedence.

Chapter 2

Device Structure and Operating Principle

2.1 The four-layer structure

Both devices are built on the classical p^+npn^+ thyristor structure, usually described as two coupled bipolar transistors: a wide-base *pnp* transistor (anode emitter, n-base, p-base) and a narrow-base *npn* transistor (cathode emitter, p-base, n-base). The gate contacts the p-base. Denoting by α_{pnp} and α_{npn} the common-base current gains, the anode current is

$$I_A = \frac{\alpha_{npn} I_G + I_{leak}}{1 - (\alpha_{pnp} + \alpha_{npn})}, \tag{2.1}$$

so the structure latches when $\alpha_{pnp} + \alpha_{npn} \to 1$ [5, 6]. In the on state, high-level injection floods the n-base with electron–hole plasma, which is the origin of the very low on-state voltage drop of these devices and, at the same time, of their slow turn-off.

2.2 The GTO thyristor

2.2.1 Cathode segmentation

To turn the device off, the *npn* transistor must be desaturated by removing holes from the p-base through the gate. Since holes must travel laterally under the cathode emitter, the cathode is not a single area but is divided into several thousand narrow fingers (typically 100 µm to 300 µm wide) interdigitated with the gate metallisation and connected in parallel. This segmentation limits the lateral distance that carriers have to travel and makes turn-off possible at all.

2.2.2 Turn-off gain and current filamentation

In a conventional GTO the negative gate current is significantly smaller than the anode current. The turn-off gain

$$\beta_{off} = \frac{I_{TGQ}}{|I_{G,peak}|} \approx \frac{\alpha_{npn}}{\alpha_{pnp} + \alpha_{npn} - 1} \tag{2.2}$$

is typically in the range 3 to 5. The device therefore turns off *in thyristor mode*: for a storage time of the order of 10 µs to 25 µs the structure remains regenerative while the plasma is progressively squeezed. Because the individual cathode fingers are not perfectly identical, the conducting area shrinks non-uniformly and the residual current concentrates in a few fingers — current filamentation — which is the dominant failure mechanism at turn-off. The maximum controllable turn-off current I_{TGQM} is set by this mechanism, not by thermal limits.

2.2.3 Consequences: snubbers and minimum times

Two practical consequences follow:

- A *turn-off (dv/dt) snubber* is mandatory. A polarised RCD snubber with a capacitor of typically 3 µF to 6 µF for a 4.5 kV/3 kA device holds the anode voltage low while the current is still being commutated, keeping the device out of the filamentation region. The snubber is bulky, lossy and expensive: its dissipation, $\frac{1}{2}CV_{DC}^2 f_{sw}$, often dominates the converter loss budget.
- *Minimum on-time and minimum off-time* of the order of 10 µs to 100 µs must be enforced by the control, because the plasma distribution must be allowed to settle before the opposite transition. This constrains the modulation strategy and effectively limits f_{sw} to a few hundred hertz.

2.2.4 Anode side variants

Three anode-side arrangements are common: *symmetric* (reverse blocking, thick n-base, high on-state drop), *asymmetric* (an n-buffer layer plus anode shorts, thinner base, lower drop, but no reverse blocking capability, used with an antiparallel diode in voltage-source converters), and *reverse conducting* (RC), where the freewheeling diode is integrated on the same wafer.

2.3 The GCT and the IGCT

2.3.1 Hard drive: unity turn-off gain

The GCT is a GTO whose gate is driven *hard*: the gate unit sinks the entire anode current, $\beta_{off} \approx 1$, in less than about one microsecond [7, 3]. The whole cathode current is commutated from the cathode into the gate *before* the anode voltage starts to rise. The *npn* transistor is then completely cut off and the device finishes turning off as an *open-base pnp transistor*: the structure is no longer regenerative, so there is no plasma pinching and no filamentation.

The essential requirement is therefore not a large gate charge but a very high gate-current slope. Since

$$\frac{\mathrm{d}i_G}{\mathrm{d}t} = \frac{V_{GK}}{L_\sigma}, \tag{2.3}$$

with the gate–cathode reverse voltage V_{GK} limited to about 20 V by the avalanche breakdown of the gate–cathode junction, the only remaining degree of freedom is the loop inductance L_σ. Achieving 6 kA/µs to 10 kA/µs requires L_σ of a few nanohenries, which is impossible with a cabled gate connection. Hence the *integration* of the gate unit with the device: the IGCT is a GCT plus a coaxially bonded, multilayer-PCB gate unit.

2.3.2 Silicon improvements

The hard-drive concept is combined with several device-level improvements [8, 6]:

- *Transparent anode emitter*: a thin, lightly doped p emitter whose injection efficiency depends on current density, replacing anode shorts; it reduces the tail current and improves the on-state/turn-off trade-off.
- *Buffer layer* (n field stop): allows a thinner n-base for a given blocking voltage, hence lower on-state voltage and lower stored charge.
- *Free-floating silicon* in a press-pack housing: double-sided cooling, high thermal cycling capability, and the important property that the device fails as a stable short circuit (short-circuit failure mode, SCFM), which allows series connection with $n + 1$ redundancy.
- *Reverse-conducting* (RC-IGCT) and *reverse-blocking* (RB-IGCT) variants, the latter used in current-source converters [9].

2.3.3 Snubberless operation

Because turn-off is homogeneous, the RCD turn-off snubber can be removed. What remains is a *di/dt clamp*: a saturable or air-core inductor of a few microhenries in the DC link branch, with its own freewheeling path, which limits the turn-on current slope of the IGCT and the reverse recovery stress of the opposing diode to about 1 kA/µs. This is a single component shared by a whole phase leg, in contrast with one snubber per switch.

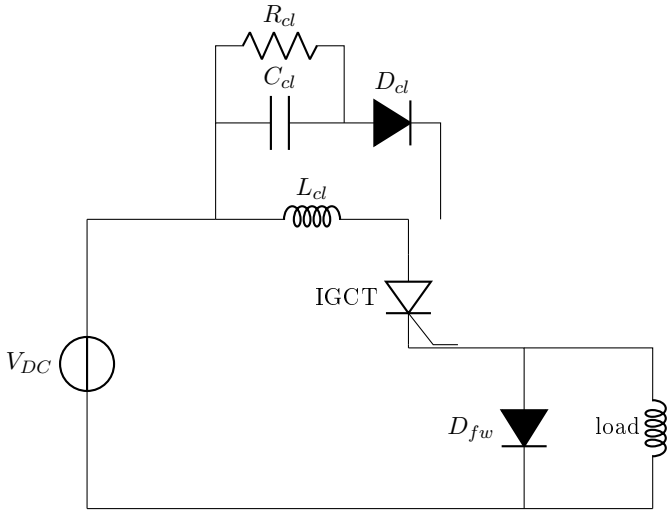


Figure 2.1: Simplified snubberless IGCT phase leg with di/dt clamp circuit (L_{cl}, D_{cl}, C_{cl}, R_{cl}). No individual RCD turn-off snubber is required.

Chapter 3

Static and Dynamic Characteristics

3.1 On-state and conduction losses

In the on state the device behaves as a p-i-n diode in series with a junction. For loss calculation the usual piecewise-linear model is adequate:

$$v_T(i_T) = V_{T0} + r_T\, i_T, \qquad P_{cond} = V_{T0}\, I_{T,avg} + r_T\, I_{T,rms}^2. \tag{3.1}$$

Typical values for a 4.5 kV device at rated current are $V_{T0} \approx 1.2\,\mathrm{V}$ to $1.7\,\mathrm{V}$ and a total on-state drop of 2 V to 3 V at several kiloamperes — markedly lower than an IGBT of the same blocking class, and the main reason these devices survive in very high power converters.

3.2 Turn-on

Turn-on is initiated by a positive gate pulse. Because the plasma spreads laterally from the gate fingers at a finite velocity, the critical rate of rise of on-state current $\mathrm{d}i/\mathrm{d}t$ must be limited externally; the clamp inductor of Fig. 2.1 performs this function. A hard, high $\mathrm{d}i_G/\mathrm{d}t$ turn-on pulse improves the initial conducting area and reduces turn-on losses and localised heating.

3.3 Turn-off

The turn-off waveform is conventionally divided into:

Storage time t_s from the gate turn-off command to the beginning of the anode voltage rise. 10 µs to 25 µs for a GTO, of the order of 1 µs for an IGCT, and much less dispersed in the latter case — which is what makes series connection practical.

Fall time t_f rapid collapse of the anode current, typically 1 µs to 2 µs.

Tail current slow recombination of the residual charge in the n-base, lasting several microseconds and contributing a substantial part of the turn-off energy, since it flows at full DC-link voltage.

The turn-off energy is obtained by integration over the whole transient:

$$E_{off} = \int_{t_0}^{t_1} v_{AK}(t)\, i_A(t)\, \mathrm{d}t. \tag{3.2}$$

3.4 Safe operating area and limits

Three distinct limits must be respected: the maximum controllable turn-off current I_{TGQM} (filamentation limit, strongly gate-unit dependent), the RBSOA in the voltage–current plane

during turn-off, and the maximum junction temperature. In addition, cosmic-ray induced single-event burnout imposes a derating of the DC-link voltage with respect to the rated blocking voltage, usually to 50 % to 60 % of V_{DRM} [10].

3.5 Comparison

Table 3.1: Representative comparison of GTO, IGCT and HV-IGBT.

	GTO	**IGCT**	**HV-IGBT**
Blocking voltage	4.5 kV to 6 kV	4.5 kV to 10 kV	3.3 kV to 6.5 kV
Turn-off current	3 kA to 6 kA	3 kA to 6 kA	0.6 kA to 1.5 kA
Turn-off gain	3 to 5	≈ 1	voltage driven
Storage time	10 µs to 25 µs	≈ 1 µs	—
Turn-off snubber	required	not required	not required
On-state drop	low	low	higher
Typical f_{sw}	200 Hz to 500 Hz	0.5 kHz to 1 kHz	1 kHz to 2 kHz
Gate drive power	high	high	low
Failure mode	short circuit (press-pack)	short circuit (press-pack)	module: open/rupture

Chapter 4

Gate Driver Design

4.1 Why the gate unit defines the device

For a MOSFET or an IGBT the gate driver delivers a charge of the order of microcoulombs and its main design problems are speed, isolation and noise immunity. For a GTO or a GCT the gate unit is a *power converter in its own right*: it must sink kiloamperes for microseconds, with a current slope of kiloamperes per microsecond, from a low-voltage rail, tens or hundreds of times per second. The gate unit therefore determines:

- the maximum controllable turn-off current;
- whether a turn-off snubber is needed;
- the spread of the storage time, and hence the feasibility of series connection;
- a non-negligible share of the converter losses and volume.

This is why an IGCT is delivered as an inseparable assembly of GCT wafer, press-pack housing and gate unit [4, 11].

4.2 Functional requirements

A complete gate unit must provide the following functions.

1. **Turn-on pulse**: a positive gate current pulse with high $\mathrm{d}i_G/\mathrm{d}t$ and a peak of tens of amperes (GTO) to a few hundred amperes (GCT), lasting a few microseconds, to ignite as much cathode area as possible simultaneously.
2. **Back-porch (on-state) current**: a continuous positive gate current maintained for the whole conduction interval, from a few amperes to a few tens of amperes depending on device size. It prevents partial de-latching at low anode current and improves $\mathrm{d}i/\mathrm{d}t$ robustness.
3. **Turn-off pulse**: a negative gate current whose amplitude equals the anode current (GCT) or a fraction of it (GTO), with the highest achievable slope.
4. **Reverse bias hold**: a sustained negative gate–cathode voltage during the whole off interval, to keep the *npn* emitter junction reverse biased and prevent spurious re-triggering by $\mathrm{d}v/\mathrm{d}t$.
5. **Auxiliary supply**: isolated for the full converter voltage, and able to deliver the average gate power.
6. **Command interface**: optical, with low and reproducible propagation delay.
7. **Monitoring and protection**: supply undervoltage lockout, minimum on/off time enforcement, status feedback to the controller.

4.3 Sizing the turn-off stage

4.3.1 Loop inductance

Equation (2.3) is the central design equation. With V_{GK} fixed by the gate–cathode avalanche voltage (about 20 V to 22 V), the required slope maps directly onto a maximum admissible loop inductance:

$$L_{\sigma,max} = \frac{V_{GK}}{(\mathrm{d}i_G/\mathrm{d}t)_{req}}. \tag{4.1}$$

For $V_{GK} = 20\,\mathrm{V}$ and $\mathrm{d}i_G/\mathrm{d}t = 6\,\mathrm{kA/\mu s}$ this gives $L_{\sigma,max} \approx 3.3\,\mathrm{nH}$. Every element of the loop — the capacitor bank ESL, the MOSFET package inductance, the PCB planes, the contact to the gate ring — must be budgeted against this figure. In a GTO gate unit, where $\beta_{off} \approx 4$ is accepted, a loop inductance of a few hundred nanohenries gives 30 A/µs to 40 A/µs, which is the practical difference between the two technologies.

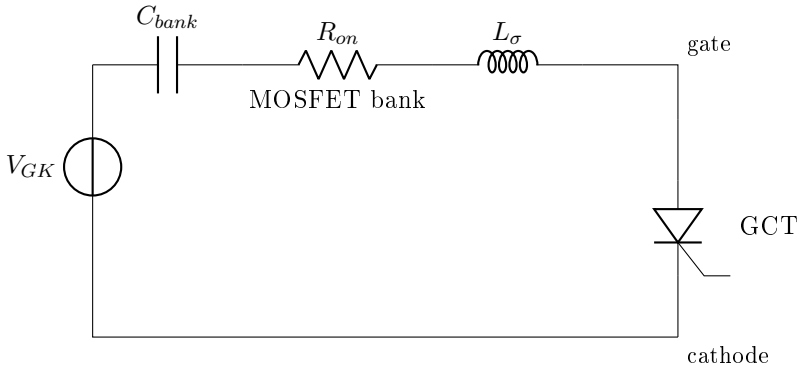


Figure 4.1: Equivalent circuit of the IGCT turn-off loop. The current slope is set by V_{GK}/L_σ; the capacitor bank must supply the full anode current with negligible voltage sag.

4.3.2 Capacitor bank

During turn-off the bank supplies the charge extracted from the device. For a triangular gate-current pulse rising to I_{TGQ} at slope S, the commutation time is $t_c = I_{TGQ}/S$ and the charge is

$$Q = \tfrac{1}{2}\, I_{TGQ}\, t_c = \frac{I_{TGQ}^2}{2S}, \qquad C_{bank} \geq \frac{Q}{\Delta V_{acc}}, \tag{4.2}$$

where ΔV_{acc} is the voltage sag that can be tolerated without degrading the slope (typically 5 % to 10 % of V_{GK}). The bank is realised as a large parallel combination of low-ESL electrolytic and film or ceramic capacitors distributed around the gate ring, so that the ESL of the bank adds in parallel and the inductance budget is met.

4.3.3 Switch bank

The turn-off switch is a parallel array of low-voltage ($\approx 30\,\mathrm{V}$), low-$R_{DS(on)}$ MOSFETs — typically several tens of devices — arranged symmetrically around the gate ring. The number N follows from two constraints:

$$N \geq \frac{I_{TGQ}}{I_{D,pulse}}, \qquad N \geq \frac{R_{DS(on)}\, I_{TGQ}}{\Delta V_R}, \tag{4.3}$$

the second ensuring that the resistive drop ΔV_R does not eat significantly into the 20 V available for the $\mathrm{d}i/\mathrm{d}t$. Symmetry matters as much as the count: an asymmetric layout produces unequal sharing and local overheating, and increases the effective loop inductance.

4.3.4 Worked example

Consider a 4.5 kV GCT with $I_{TGQ} = 4\,\mathrm{kA}$, $V_{GK} = 20\,\mathrm{V}$ and a required $S = 6\,\mathrm{kA/\mu s}$.

- Loop inductance: $L_\sigma \leq 20/6 \times 10^9 = 3.3\,\mathrm{nH}$.
- Commutation time: $t_c = 4000/6 \times 10^9 = 0.67\,\mu\mathrm{s}$.
- Extracted charge: $Q = \frac{1}{2} \cdot 4000 \cdot 0.67 \times 10^{-6} \approx 1.3\,\mathrm{mC}$.
- Capacitor bank for $\Delta V_{acc} = 2\,\mathrm{V}$: $C_{bank} \geq 1.3 \times 10^{-3}/2 \approx 670\,\mu\mathrm{F}$.
- MOSFET count with $I_{D,pulse} = 100\,\mathrm{A}$ per device: $N \geq 40$; choosing $N = 60$ with $R_{DS(on)} = 4\,\mathrm{m\Omega}$ gives a bank resistance of $67\,\mu\Omega$ and $\Delta V_R = 0.27\,\mathrm{V}$, which is negligible.
- Energy drawn from the bank per turn-off: $E \approx \frac{1}{2} L_\sigma I_{TGQ}^2 + Q\,V_{GK} \approx 26\,\mathrm{mJ} + 26\,\mathrm{mJ}$, so at 500 Hz the turn-off contribution to the gate-unit power is of the order of a few tens of watts.

The result — a few nanohenries, hundreds of microfarads, tens of MOSFETs — explains the physical form of an IGCT gate unit: an annular, multilayer printed circuit board of 15 cm to 25 cm diameter, bolted concentrically to the press-pack gate ring, with the return path immediately underneath the outgoing path.

4.4 The GTO gate unit

Because the GTO operates with $\beta_{off} > 1$, the switch and capacitor bank are smaller, but three additional problems appear.

1. **Storage time control.** The long and dispersed storage time must be accounted for in the dead-time of the modulator, and cannot be compensated by the gate unit.
2. **Coordination with the snubber.** The turn-off snubber must be correctly sized and low-inductance; the gate unit alone cannot guarantee the RBSOA.
3. **Interlocks.** Minimum on-time and minimum off-time must be enforced in hardware inside the gate unit, not only in the modulator, because a violation during a fault sequence destroys the device.

A classical GTO gate unit uses three separate paths sharing the gate terminal: a positive pulse path (charged capacitor plus switch and series inductor to shape the pulse), a back-porch path (a current-limited linear or switched source), and the negative path (bank plus MOSFET array) with a hold-off supply for the reverse bias.

4.5 Auxiliary power supply

The gate unit power consumption has three components:

$$P_{GU} = \underbrace{V_{bp} I_{bp} D}_{\text{back porch}} + \underbrace{(E_{on}^G + E_{off}^G) f_{sw}}_{\text{switching}} + \underbrace{P_{aux}}_{\text{logic, monitoring}}, \tag{4.4}$$

and typically ranges from a few tens of watts to a few hundred watts per device. Two supply strategies are used:

- An isolated DC/DC converter fed from a low-voltage bus through a high-isolation transformer (partial discharge and creepage rated for the full converter voltage, plus the transient $\mathrm{d}v/\mathrm{d}t$ common-mode stress of the switching leg).
- Self-supply from the device itself, harvesting energy from the anode voltage or from the clamp circuit during the off interval, with a start-up path.

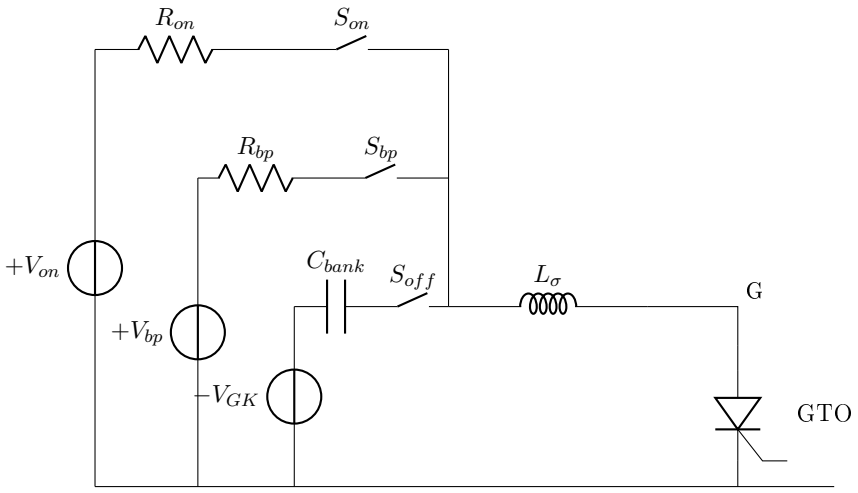


Figure 4.2: Conceptual three-path GTO gate unit: turn-on pulse, back-porch source and turn-off bank sharing the gate terminal.

In both cases the gate unit stores enough energy locally to perform at least one safe turn-off after a supply failure, and signals the loss of supply to the controller.

4.6 Signal interface, isolation and synchronisation

The command is transmitted over plastic or glass optical fibre. Two requirements dominate:

- **Low and reproducible latency.** In series-connected stacks, the jitter between gate units directly translates into unequal voltage sharing during the transient. Fibre length matching and equal-latency receivers are standard practice.
- **Status feedback.** A return fibre reports supply status, gate-unit readiness and, in advanced units, an indication derived from the gate–cathode voltage that the device actually turned off. This feedback is what allows a controller to detect a failed device in a redundant series stack.

The common-mode transient immunity of the isolation barrier must be specified for the $\mathrm{d}v/\mathrm{d}t$ seen at the device terminals, which for an IGCT phase leg is of the order of $1\,\mathrm{kV/\mu s}$ to $5\,\mathrm{kV/\mu s}$.

4.7 Layout guidelines

- Build the gate unit as an annular multilayer PCB coaxial with the press-pack; keep the go and return conductors on adjacent layers with minimum dielectric thickness to minimise L_{σ}.
- Distribute the capacitor bank and the MOSFETs uniformly around the ring; each MOSFET should see nominally the same path length.
- Keep the gate-unit ground reference tied to the device cathode, and treat every measurement referenced to it as floating at full converter potential.
- Provide thermal management for the MOSFET bank and the bank ESR: at several hundred hertz these components dissipate continuously.
- Route the optical receivers and the logic supply away from the pulse loop; the $\mathrm{d}i/\mathrm{d}t$ in the turn-off loop is the largest interference source in the whole converter.

4.8 Protection and interlocks

- **Undervoltage lockout** on the gate-unit rails, with inhibition of turn-on commands.

- **Minimum on-time / minimum off-time** enforced in hardware.
- **Turn-off inhibition window** during the storage phase, so that a command reversal cannot interrupt an ongoing transition.
- **Overcurrent handling.** A GCT cannot turn off an arbitrary fault current: above I_{TGQM} the device will fail. Short-circuit protection is therefore performed at converter level (fault current limiting by the clamp inductor, crowbar or fuse coordination), not by the gate unit alone.
- **Failure reporting** through the return fibre, allowing the controller to trip the stack or, in redundant configurations, to continue operating with the failed device shorted.

Chapter 5

Applications and Perspective

5.1 Typical applications

GTOs dominated high-power converters from the 1980s until the late 1990s, in traction drives, rolling mill drives and early FACTS equipment. IGCTs replaced them in most new designs from the late 1990s onwards [12, 1]. Present application areas include:

- Medium-voltage industrial drives, typically three-level neutral-point clamped inverters in the 1 MW to 30 MW range [2, 13];
- Marine propulsion and mine hoists;
- Grid interfaces: STATCOM, back-to-back interties, wind-farm converters;
- Pumped storage and large excitation systems;
- DC circuit breakers and fault current limiters, where the very high surge and turn-off current capability is decisive.

5.2 Position with respect to IGBT and wide-bandgap devices

The high-voltage IGBT has taken over a large part of the medium-voltage market thanks to its simple voltage-controlled gate, higher switching frequency and mature module technology. The IGCT retains an advantage where conduction losses dominate, where the surge current capability and the short-circuit failure mode are required, and in press-pack series stacks. Wide-bandgap devices — SiC MOSFETs and SiC IGBTs at 3.3 kV to 10 kV — are progressively addressing the same voltage classes with much higher switching frequency, but not yet with comparable single-die current ratings or surge capability.

5.3 Conclusion

The GTO and the IGCT differ far less in silicon than in the way the gate is driven. Hard drive — unity turn-off gain achieved through a gate loop of a few nanohenries — transforms a filamentation-limited, snubber-dependent switch into a homogeneous, snubberless one, with a storage time reduced by an order of magnitude and a dispersion low enough to allow series connection. Any design work on these devices is therefore, to a large extent, gate-unit design work: loop inductance budgeting, capacitor and switch bank sizing, auxiliary supply energy management, and interlock logic.

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